

System Center Operations Manager (SCOM) – Detailed Features and Capabilities

Introduction

System Center Operations Manager (SCOM) is an enterprise monitoring solution developed by Microsoft. It is part of the Microsoft System Center suite. SCOM is used by organizations to monitor the health, performance, and availability of IT infrastructure such as servers, applications, databases, and network devices. It provides centralized control and helps administrators detect and resolve issues before they affect users.

1. Infrastructure Monitoring

SCOM provides continuous monitoring of physical and virtual infrastructure.

- It monitors Windows and Linux servers by installing agents on them.
- Tracks performance metrics like CPU usage, memory utilization, disk space, disk I/O, and network traffic.
- Detects hardware failures, service stoppages, and system crashes.
- Monitors virtual environments such as Microsoft Hyper-V to ensure virtual machines are running efficiently.

This ensures high system availability and prevents downtime.

2. Application Performance Monitoring (APM)

SCOM monitors enterprise applications to ensure they are functioning correctly.

- Tracks application response times and user transactions.
- Detects exceptions and failures in .NET applications.
- Identifies slow database queries.
- Performs root cause analysis to determine where the problem occurred.

For example, it provides deep monitoring for applications like Microsoft SQL Server and Microsoft Exchange Server by checking database health, mail flow, and service availability.

This helps maintain smooth business operations.

3. Management Packs

Management Packs are pre-configured sets of monitoring rules.

- They define what components to monitor.
- They specify thresholds for alerts.
- They include reports and performance views.
- They contain knowledge articles for troubleshooting.

Microsoft provides management packs for products like Active Directory, enabling automatic monitoring of domain controllers, replication, and authentication services.

Custom management packs can also be created for third-party or in-house applications.

4. Real-Time Alerts and Notifications

SCOM generates alerts automatically when issues occur.

- If CPU usage exceeds a defined threshold, an alert is triggered.
- If a service stops unexpectedly, administrators are notified.
- Alerts are categorized by severity (Information, Warning, Critical).
- Notifications can be sent via email or integrated into helpdesk/ticketing systems.

This enables proactive issue resolution before users are affected.

5. Reporting and Dashboards

SCOM provides detailed reporting capabilities.

- Real-time dashboards display system health status.
- Performance trend reports show usage patterns over time.
- SLA (Service Level Agreement) reports measure service availability.
- Historical data is stored in a Data Warehouse database for long-term analysis.

Reports help management make informed IT decisions.

6. Distributed Application Monitoring

SCOM can monitor multi-tier applications across different servers.

For example, a web application may consist of:

- A web server
- An application server
- A database server

SCOM maps these dependencies and shows how they are connected. If the database fails, SCOM identifies it as the root cause rather than just showing the web server as down.

This improves troubleshooting efficiency.

7. Cloud and Hybrid Monitoring

SCOM supports hybrid IT environments.

- Monitors on-premises servers.
- Integrates with Microsoft Azure for cloud monitoring.
- Supports monitoring of Azure virtual machines and services.
- Enables centralized management of both cloud and local infrastructure.

This is important for modern organizations using cloud technology.

8. Automation and Integration

SCOM reduces manual work through automation.

- Supports PowerShell scripting for automated tasks.

- Integrates with System Center Orchestrator for workflow automation.
- Can connect with IT Service Management (ITSM) tools to create tickets automatically.

Automation improves efficiency and reduces human errors.

9. Scalability and Security

SCOM is designed for large enterprises.

- Can monitor thousands of devices.
- Supports distributed management servers.
- Provides role-based access control (RBAC) for security.
- Supports high availability configurations.

This makes it suitable for large organizations and data centers.

10. Advantages and Limitations

Advantages:

- Centralized monitoring platform
- Deep integration with Microsoft products
- Proactive problem detection
- Detailed reporting and analytics
- Enterprise scalability

Limitations:

- Complex installation and configuration
- Requires trained administrators
- Licensing cost may be high
- Resource-intensive in large environments

Operations Manager Basics

Microsoft System Center Operations Manager (SCOM) is an enterprise monitoring system used by organizations to monitor servers, applications, and network devices. It collects data from many computers and helps IT administrators detect problems, analyze performance, and generate reports.

1. Management Server

The Management Server is the core component of Operations Manager.

Functions

- Receives monitoring data from agents installed on servers and devices.
- Processes alerts, events, and performance data.
- Applies management packs (rules and monitoring settings).
- Sends alerts or notifications when problems occur.

Example

If a server's CPU usage becomes too high, the agent sends data to the Management Server, which analyzes it and generates an alert.

Key Role

Acts as the central controller of the monitoring system.

2. Web Console Server

The Web Console Server provides a web-based interface for users to monitor systems.

Functions

- Allows administrators to view alerts, health status, and performance data through a web browser.
- Enables remote monitoring without installing the full console.
- Provides dashboards and system status views.

Example

An administrator can open a browser and check server health from anywhere within the network.

Key Role

Provides remote access to monitoring information.

3. Reporting Server

The Reporting Server generates detailed reports using collected monitoring data.

Functions

- Creates reports about system performance, availability, and trends.
- Uses SQL Server Reporting Services (SSRS).

- Helps organizations analyze historical data.

Example Reports

- Server uptime reports
- CPU and memory usage trends
- Network performance reports

Key Role

Converts monitoring data into useful reports for analysis and decision-making.

4. Operational Database

The Operational Database stores current monitoring data.

Characteristics

- Keeps recent alerts, events, and performance data.
- Data is stored for a short period (usually days or weeks).
- Used by the Management Server for real-time monitoring.

Example

When a disk failure alert occurs, it is stored in the Operational Database so administrators can view it immediately.

Key Role

Stores live and recent monitoring information.

5. Data Warehouse Database

The Data Warehouse Database stores long-term historical data.

Characteristics

- Keeps monitoring data for months or years.
- Used mainly for reporting and trend analysis.
- Data is transferred from the Operational Database to the Data Warehouse.

Example

Organizations analyze yearly server performance trends using data stored in the Data Warehouse.

Key Role

Stores historical monitoring data for reports and analysis.

Relationship Between Components

All components work together in the following flow:

1. Agents collect data from servers and devices.
2. Data is sent to the Management Server.
3. Data is stored in the Operational Database for current monitoring.

4. Data is transferred to the Data Warehouse for long-term storage.
5. Users access monitoring data through the Web Console.
6. The Reporting Server generates reports using the Data Warehouse data.

Install Agent on Windows Using the Discovery Wizard

Deploying clients (agents) is an important task in SCOM. The Operations Manager Agent is a small software installed on computers that collects monitoring data such as events, performance information, and alerts, and sends it to the Management Server.

One of the easiest ways to install agents on Windows computers is by using the Discovery Wizard in the Operations Manager Console.

1. Introduction

In SCOM, monitoring is done through agents installed on target computers. These agents communicate with the Management Server and send information about the health and performance of the system.

The Discovery Wizard helps administrators automatically find computers in the network and install agents remotely.

2. Requirements Before Installing Agents

Before using the Discovery Wizard, the following requirements must be met:

- The target computer must be running Windows.
- The computer must be connected to the network.
- Administrator credentials must be available.
- Firewall settings must allow remote installation.
- The computer should be part of the same domain or trusted domain.

3. Steps to Install Agent Using Discovery Wizard

Step 1: Open Operations Manager Console

- Log in to the SCOM Management Server.
- Open the Operations Manager Console.

Step 2: Start the Discovery Wizard

- Go to the Administration workspace.
- Select Device Management.
- Click Discovery Wizard to begin the process.

Step 3: Select Discovery Type

Two discovery options are available:

1. Automatic Discovery – Scans the entire domain for computers.
2. Advanced Discovery – Allows selecting specific computers.

Usually Advanced Discovery is used to install agents on selected systems.

Step 4: Specify Discovery Criteria

- Enter the computer names or browse Active Directory.
- Select the systems where the agent should be installed.

Step 5: Provide Administrator Credentials

- Enter domain administrator username and password.
- These credentials allow SCOM to remotely install the agent.

Step 6: Select Management Server

- Choose the Management Server that will manage the discovered computers.

Step 7: Install the Agent

- The wizard will start installing the Operations Manager Agent on the selected computers.
- Installation occurs remotely from the management server.

Step 8: Verify Installation

- After installation, the computer appears under Agent Managed in the console.
- The agent begins sending monitoring data to the management server.

4. Advantages of Using Discovery Wizard

- Easy and automatic agent deployment
- Saves time and manual effort
- Allows centralized management
- Supports multiple computer installations at once

5. Conclusion

The Discovery Wizard in SCOM provides a simple and efficient method for deploying monitoring agents on Windows computers. By installing agents, administrators can collect system information, monitor performance, detect issues, and ensure the overall health of the IT infrastructure.

Management Packs in Microsoft System Center Operations Manager (SCOM)

Introduction

A Management Pack (MP) is a collection of rules, monitors, tasks, knowledge, and configuration settings used by SCOM to monitor applications, services, operating systems, and network devices.

When a management pack is imported into a management group, SCOM automatically starts monitoring the specified components based on the default configurations and thresholds defined in that management pack.

Management packs simplify monitoring because administrators do not need to configure monitoring rules manually.

Components (Parts) of a Management Pack

A management pack can contain several components that define how monitoring should be performed.

1. Monitors

Monitors track the health state of system components.

They observe different aspects of a system such as:

- Disk usage
- CPU utilization
- Application availability
- Network status

Monitors usually have health states:

State	Meaning
Green	Healthy condition

Yellow	Warning condition
Red	Critical condition

Example:

A disk utilization monitor may show:

- Green if disk usage is below 75%
- Yellow if usage is between 75–90%
- Red if usage exceeds 90%

Monitors help administrators detect issues before they become major failures.

2. Rules

Rules define what data should be collected and what action should be taken.

Rules are commonly used to:

- Collect performance data
- Record event logs
- Generate alerts
- Run scripts

Rules follow a simple IF–THEN logic.

Example:

- IF application stops THEN generate alert
- IF login fails THEN record event

Rules mainly focus on data collection, while monitors focus on health state monitoring.

3. Tasks

Tasks are actions that administrators can run on monitored systems.

Tasks can perform operations such as:

- Restarting services
- Running scripts
- Checking disk errors
- Clearing logs

Example:

- Restart a failed application
- Run disk check command (chkdsk)

Types of Tasks

1. Diagnostic Tasks – Identify the cause of a problem

2. Recovery Tasks – Attempt to fix the problem automatically

Tasks may run manually or automatically.

4. Knowledge

Knowledge provides troubleshooting information about alerts.

It helps operators understand:

- Why the problem occurred
- How to fix it

Knowledge appears in the console as text instructions.

Types of Knowledge

Product Knowledge

- Provided by the management pack creator.

Company Knowledge

- Added by administrators with organization-specific solutions.

Example:

An alert might include steps like:

1. Check service status
2. Restart the service
3. Verify network connection

5. Views and Dashboards

Views

Views display monitoring data in the Operations Manager Console.

Common views include:

- State View
- Alerts View
- Performance View

Dashboards

Dashboards provide graphical representations such as charts and graphs to visualize system status.

They help administrators quickly understand system health.

6. Reports

Reports provide historical monitoring information.

Reports help administrators analyze:

- System performance
- Server availability
- Resource usage

Examples:

- Monthly CPU usage report
- Server uptime report

Reports use data stored in the Data Warehouse database.

7. Object Discoveries

Object discoveries identify systems or components that need monitoring.

They search the network using methods such as:

- Windows Registry
- WMI
- LDAP
- Scripts
- Database queries

Example:

Discovering all web servers or database servers in a network.

If unwanted objects are discovered, administrators can restrict monitoring using overrides.

8. Run As Profiles

Run As profiles allow monitoring tasks to run using specific user credentials.

Some rules, monitors, or tasks require special permissions.

Run As profiles allow administrators to:

- Create Run As accounts
- Assign credentials
- Specify target computers

Example:

Monitoring a database may require database administrator credentials.

Sealed and Unsealed Management Packs

Sealed Management Pack

- Cannot be modified
- Provided by vendors
- File extension: .mp

Example: Vendor-provided application management packs.

Unsealed Management Pack

- Can be edited

- Used for custom configurations
- File extension: .xml

Administrators usually store custom settings and overrides in unsealed management packs.

Management Pack Dependencies

Some management packs depend on other management packs.

This is called dependency.

Example:

An application management pack may require a Windows Server management pack.

If required management packs are missing, the import process will fail.

Administrators can check dependencies in the Operations Manager Console.

Advantages of Management Packs

- Provide automatic monitoring configuration
- Reduce manual setup
- Detect issues early
- Provide troubleshooting guidance
- Improve system reliability
- Enable efficient IT infrastructure monitoring

Management Packs Installed with Microsoft System Center

Operations Manager

Introduction

When Operations Manager is installed, several default management packs are automatically installed. These management packs provide the basic monitoring framework required for managing computers, applications, and services in the environment.

These default management packs contain classes, rules, monitors, discoveries, and views that help Operations Manager monitor system health and performance.

They also act as foundation or library management packs on which other management packs depend.

Purpose of Default Management Packs

The management packs installed with Operations Manager help to:

- Provide basic monitoring functionality

- Define core classes and objects
- Support alerting and event collection
- Enable performance monitoring
- Provide views and reports
- Act as libraries for other management packs

Common Management Packs Installed with Operations Manager

1. System Library Management Pack

The System Library Management Pack defines the basic classes used in Operations Manager.

Functions:

- Defines fundamental objects such as computer, operating system, and application.
- Provides the base structure used by other management packs.

This management pack is required because many other management packs depend on it.

2. Health Library Management Pack

The Health Library Management Pack defines the health model used in monitoring.

Functions:

- Defines health states such as:
 - o Healthy (Green)
 - o Warning (Yellow)
 - o Critical (Red)
- Provides monitoring structure for system components.

This pack allows SCOM to determine whether a system is healthy or experiencing problems.

3. Performance Library Management Pack

The Performance Library Management Pack manages performance data collection.

Functions:

- Collects performance metrics such as:
 - o CPU usage
 - o Memory usage
 - o Disk activity
- Stores performance information for analysis.

This helps administrators analyze system performance and detect bottlenecks.

4. Alert Management Pack

The Alert Management Pack manages alerts generated by monitors and rules.

Functions:

- Generates alerts when problems occur.
- Displays alerts in the Operations Manager Console.
- Helps administrators identify system issues quickly.

Example:

If a service stops running, an alert is generated and displayed in the console.

5. Microsoft Windows Library Management Pack

This management pack defines Windows-specific monitoring classes.

Functions:

- Supports monitoring of Windows-based systems
- Defines components such as:
 - o Windows computers
 - o Windows services
 - o Windows processes

Many Windows application management packs depend on this library.

6. Operations Manager Internal Library Management Pack

This pack is used for internal operations of Operations Manager.

Functions:

- Supports internal workflows
- Maintains communication between components
- Ensures smooth operation of monitoring processes

7. Reports Library Management Pack

The Reports Library Management Pack supports report generation.

Functions:

- Defines reporting components
- Helps create reports based on collected monitoring data

Examples of reports:

- System availability report
- Performance trend report

8. Service Level Tracking Management Pack

This pack supports Service Level Monitoring (SLM).

Functions:

- Tracks service availability
- Measures service performance
- Helps organizations maintain Service Level Agreements (SLAs)

Example:

Tracking uptime of a business-critical application.

Importance of These Management Packs

The management packs installed with Operations Manager are important because they:

- Provide basic monitoring functionality
- Support application and server monitoring
- Serve as foundation packs for other management packs
- Enable alerting, reporting, and performance monitoring

Without these packs, Operations Manager cannot perform monitoring tasks effectively.

Conclusion

Management packs installed with Operations Manager form the core monitoring framework of the system. They define basic classes, health models, performance monitoring, alerts, and reporting capabilities. These packs also serve as libraries for additional management packs, enabling Operations Manager to monitor a wide variety of applications and systems.